

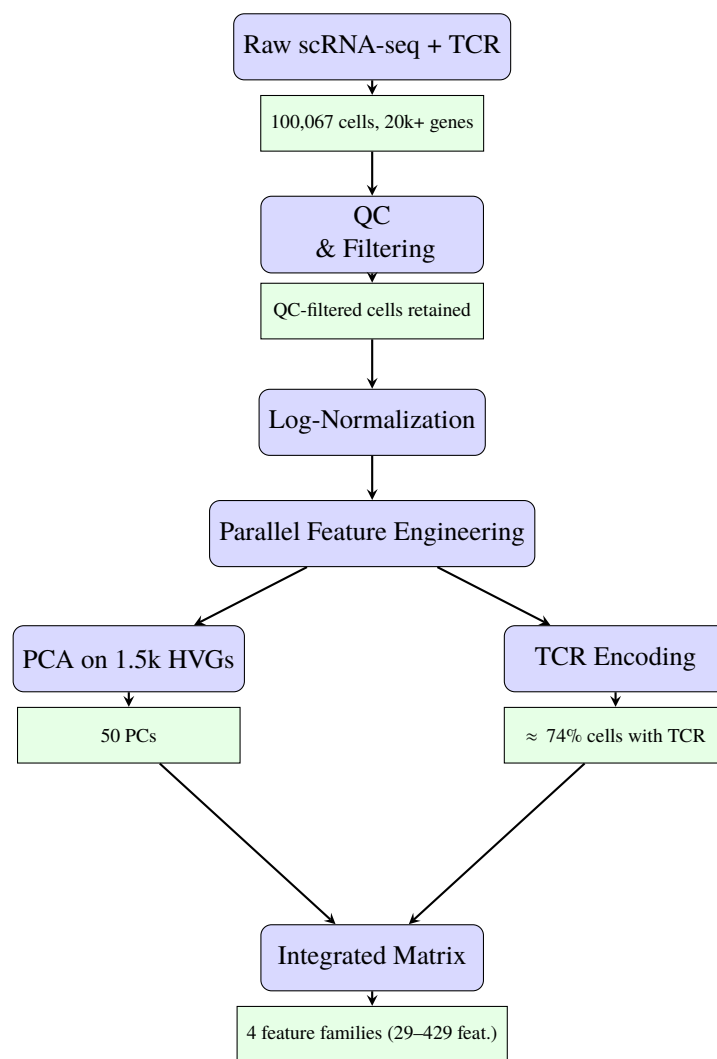


Figure 1: Enhanced Data Processing Pipeline with quantitative details at each stage, showing cell counts, feature dimensions, and variance explained.

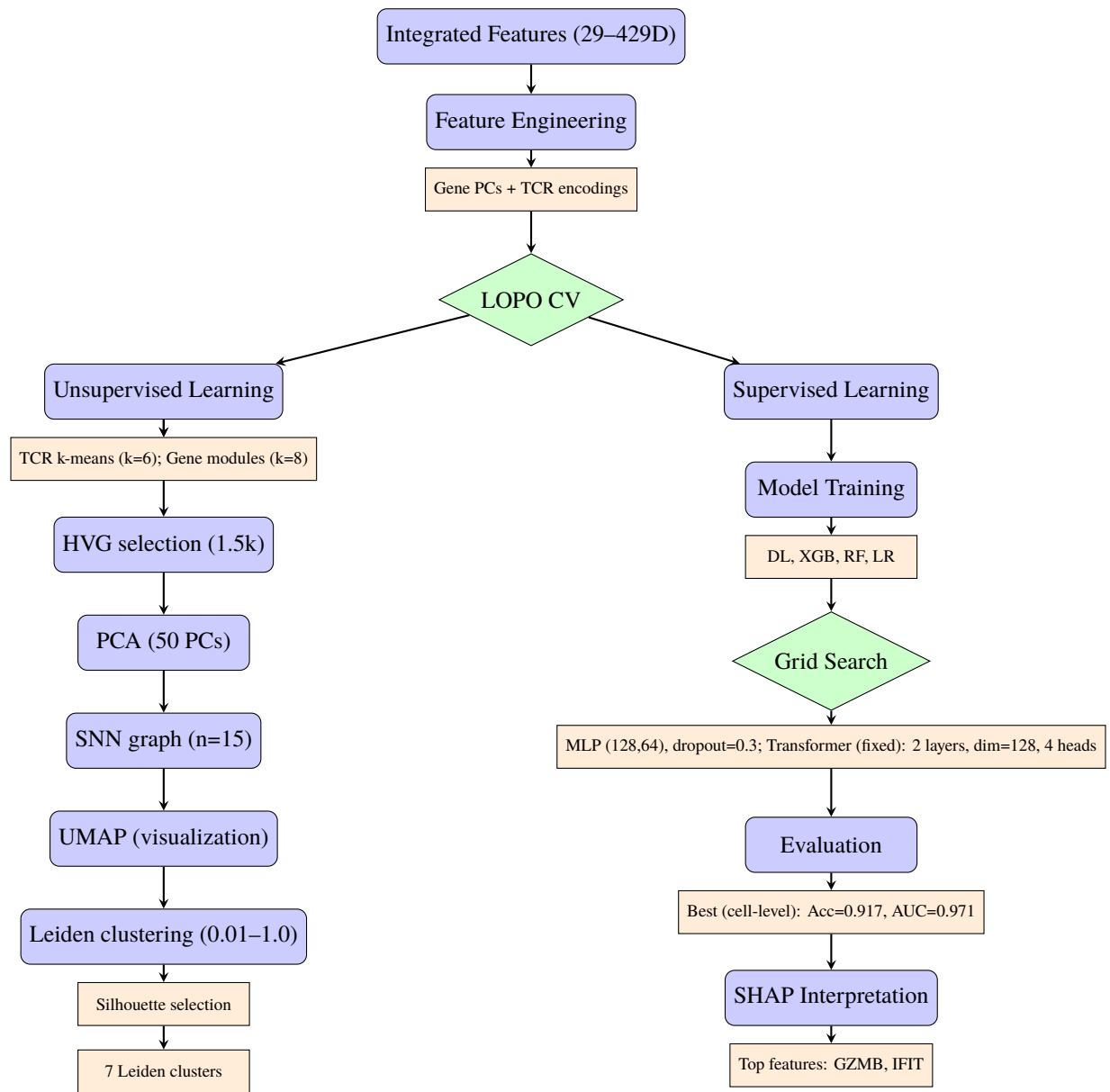
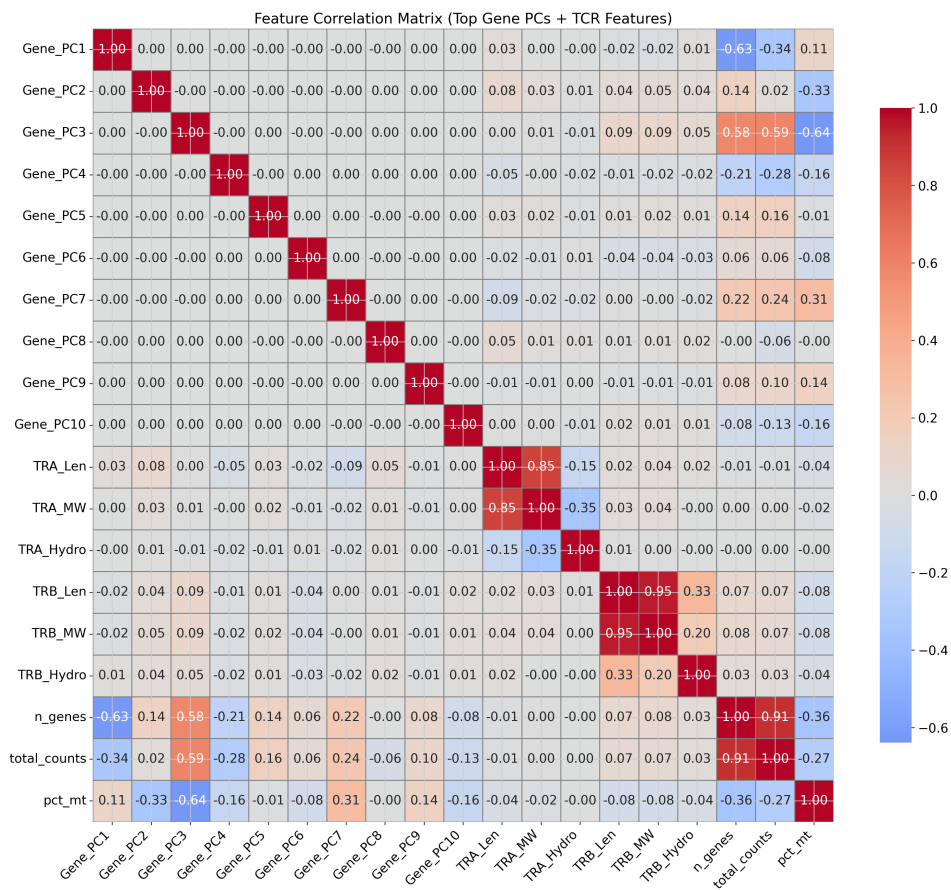
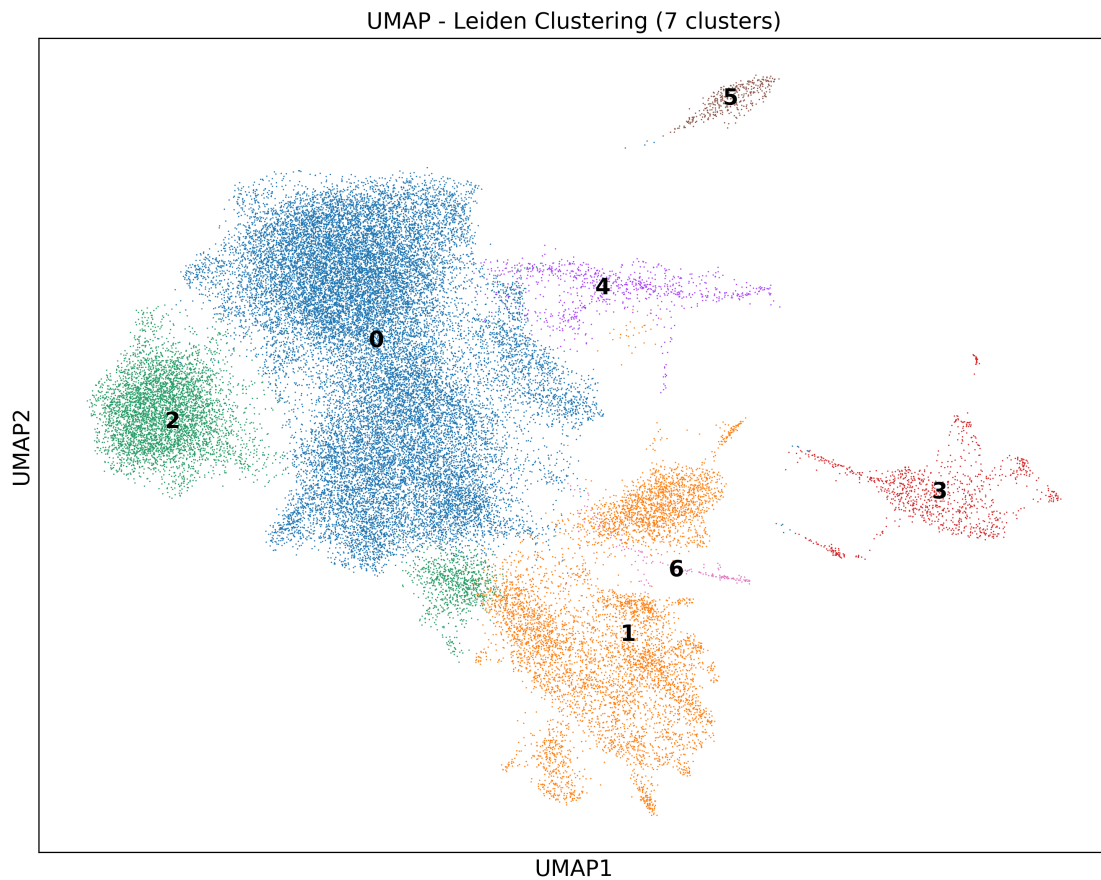


Figure 2: Enhanced Modeling Pipeline: Two-stage strategy with feature engineering, hyperparameter optimization via grid/randomized search, and LOPO cross-validation. Representative deep model hyperparameters (MLP selected via inner CV; CNN/BiLSTM/Transformer used fixed representative configurations) and Leiden clustering outputs (target: 7 clusters, silhouette-optimized resolution) are shown.



(a) Heatmap of the correlation matrix for the top engineered features (first 20 PCs and select receptor summaries).



(b) UMAP visualization of Leiden clusters colored by cluster assignment.

Figure 3: Heatmap and UMAP visualization of engineered features and clusters.